Lecture 9 Tutorial — Performance Evaluation

Q1. Confusion Matrix Entries (Fill in the blanks)

For binary classification, fill in the correct term (TP / TN / FP / FN) for each description:

1. Predicted positive, actually positive: ____________
2. Predicted positive, actually negative: ____________
3. Predicted negative, actually positive: ____________
4. Predicted negative, actually negative: ____________

Q2. Metrics from Confusion Matrix (Formula Completion)

Complete each definition using (TP, FP, FN, TN):

1. Accuracy:
$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN}$$
2. Recall (Sensitivity / TPR):
$$\text{Recall} = \frac{\_\_\_\_\_\_\_\_}{\_\_\_\_\_\_\_\_\_\_\_\_}$$
3. Precision:
$$\text{Precision} = \frac{\_\_\_\_\_\_\_\_}{\_\_\_\_\_\_\_\_\_\_\_\_}$$
4. Specificity (TNR):
$$\text{Specificity} = \frac{\_\_\_\_\_\_\_\_}{\_\_\_\_\_\_\_\_\_\_\_\_}$$
5. Fall-out (FPR):
$$\text{Fallout} = \frac{\_\_\_\_\_\_\_\_}{\_\_\_\_\_\_\_\_\_\_\_\_}$$
6. Miss rate (FNR):
$$\text{Miss} = \frac{\_\_\_\_\_\_\_\_}{\_\_\_\_\_\_\_\_\_\_\_\_}$$

Q3. Complementary Rates (Short Proof)

Using the lecture formulas, prove each identity in one line:

1. Recall + Miss = 1
2. Specificity + Fallout = 1

Q4. Compute Metrics for Given Cases (Calculation)

For each case below, assume the order is (TP, FP, FN, TN). Compute: Accuracy, Specificity, Recall, Precision, and F_1.

Use:

$$F_1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$

(Report each metric to 3 decimal places.)

Case A: $(63, 28, 37, 72)$

Case B: $(77, 77, 23, 23)$

Case C: $(24, 88, 76, 12)$

Case D: $(76, 12, 24, 88)$

Q5. F1-score Equivalence (Derivation)

Show that the following two expressions for F_1 are equivalent:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F_1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$

(Algebraic steps only; no narrative.)

Q6. Thresholding a Score-Based Classifier (Confusion Matrix + Metrics)

You are given predicted scores and ground-truth labels $(1 = \text{positive}, 0 = \text{negative})$:

i	score	label
1	0.11	1
2	0.35	0
3	0.92	0
4	0.51	1
5	0.49	0
6	0.81	1
7	0.26	0
8	0.63	0
9	0.72	1
10	0.25	1

Decision rule: predict positive iff score $\geq \tau$.

1. For $\tau = 0.50$, compute (TP, FP, FN, TN), then compute Accuracy, Recall, Precision, and F_1.
2. For $\tau = 0.70$, compute the same quantities.
3. Which threshold gives the higher F_1? $(\tau = 0.50 \ / \ \tau = 0.70)$

Q7. ROC Operating Points (TPR/FPR Table)

Using the same data and rule in Q6, compute (FPR, TPR) for each threshold:

$$\tau \in \{0.90, 0.70, 0.50, 0.30\}.$$

Return your answer as a 4-row table:

τ	FPR	TPR
______	________	________
______	________	________
______	________	________
______	________	________

(Compute exact fractions or decimals to 3 d.p.)

Q8. AUC (Concept Check — Multiple Choice)

Which statement best matches the lecture description?

A. AUC is the area under the ROC curve.

B. AUC depends on the choice of a single threshold only.

C. AUC is computed from training accuracy.

D. AUC is a measure used only for multi-class problems.

Q9. Cross-Validation Methods (Matching)

Match each method to its description.

Methods:

A. Random sampling (hold-out)

B. Bootstrap

C. k-fold cross validation

D. Leave-one-out cross validation

Descriptions:

1. Resample with replacement from the original dataset to form training sets.
2. Partition data into two disjoint sets under an assumption about distribution uniformity.
3. Split data into mutually exclusive folds; each iteration uses one fold as test and the rest as training; average the measures.
4. Special case where the number of folds equals the number of samples.

Write: A–____, B–____, C–____, D–____.

Q10. k-fold Mechanics (Computation)

A dataset has $n = 103$ samples and you use $k = 10$ folds.

1. How many models are trained in total?
2. What are the possible fold sizes (how many folds of size 10 vs 11)?
3. If the metric values across folds are
$$m_1, \ldots, m_{10},$$
write the final reported performance in one line (symbolic expression only).

Q11. Three-way Split and Leakage (Multiple Choice)

Choose the single best answer.

Why separate the validation set from the test set?

A. Because the validation set must never be used to tune parameters.

B. Because using the validation set to select the final model biases the error estimate downward; the test set is reserved for final assessment only.

C. Because the test set is used to fit the parameters of the classifier.

D. Because the training set is only used for evaluation.

Q12. Model Selection Criteria: AIC and BIC (Calculation + Decision)

Two candidate models are evaluated on the same test dataset with size $n = 1000$. Their log-likelihoods and parameter counts are:

$$\text{Model 1:} \qquad k = 5,\ \ln(L) = -1200$$

$$\text{Model 2:} \qquad k = 12,\ \ln(L) = -1180$$

Using the lecture formulas:

$$AIC = 2k - 2\ln(L), \qquad BIC = k\ln(n) - 2\ln(L)$$

1. Compute AIC_1, AIC_2.
2. Compute BIC_1, BIC_2.
3. Under each criterion (AIC and BIC), which model is preferred? (Lower is better.)